ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 2 - Conduction, Doping & PN Junction

9 June 2026

Today's Agenda — Lecture Notes pp. 16–27

- **Part 1 — Current in Semiconductors**
  - Conduction electrons, holes, the electron–hole pair, and recombination
  - Electron current, hole current, and the contrast with a metal
- **Part 2 — N-Type & P-Type Semiconductors**
  - Doping: pentavalent donors and trivalent acceptors
  - Majority and minority carriers
- **Part 3 — The PN Junction**
  - Diffusion, the depletion region, and the barrier potential
  - The energy diagram and the energy hill
- **Part 4 — Wrap-Up**
  - Synthesis, key takeaways, and a look ahead to the diode

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
  - Explain how current is produced in a semiconductor — conduction electrons, holes, electron–hole pairs, and recombination
  - Distinguish electron current from hole current, and contrast both with conduction in a metal
  - Define doping and explain why intrinsic silicon must be modified to be useful
  - Explain how pentavalent (donor) and trivalent (acceptor) atoms create n-type and p-type material
  - Identify the majority and minority carriers in n-type and p-type material
  - Describe how a pn junction forms — diffusion, the depletion region, and the barrier potential ($\approx$ 0.7 V Si, 0.3 V Ge)
  - Interpret the energy diagram of a pn junction and explain the energy hill

Part 1 — Current in Semiconductors

Last Week — Intrinsic Silicon

- **The Pure Silicon Crystal**
  - Every silicon atom shares four covalent bonds with its neighbours — a stable crystal lattice
  - Valence electrons sit in the valence band; the conduction band above is empty
  - A band gap (~1.1 eV for silicon) separates the two bands
- **At Absolute Zero (0 K)**
  - Every valence electron is locked in a bond — there are no free carriers, so no current

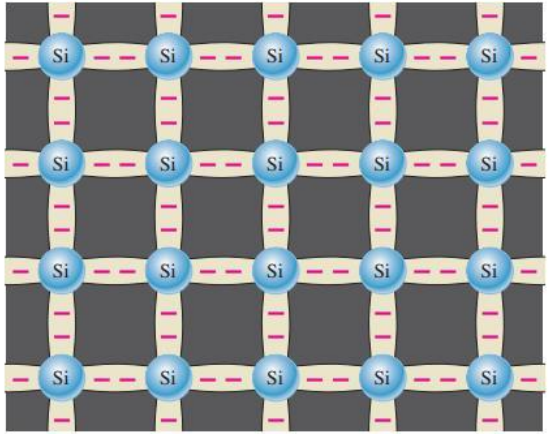


Fig 1.11 — A silicon crystal lattice

This Week — From Pure Silicon to the Diode

- **Where We Start**
  - Intrinsic silicon at 0 K: every bond intact, no free carriers (see figure)
- **What We Build Today**
  - Add heat → a few carriers appear and silicon conducts weakly
  - Add impurities (doping) → set the conductivity by design
  - Join p-type to n-type → the first electronic device, the diode

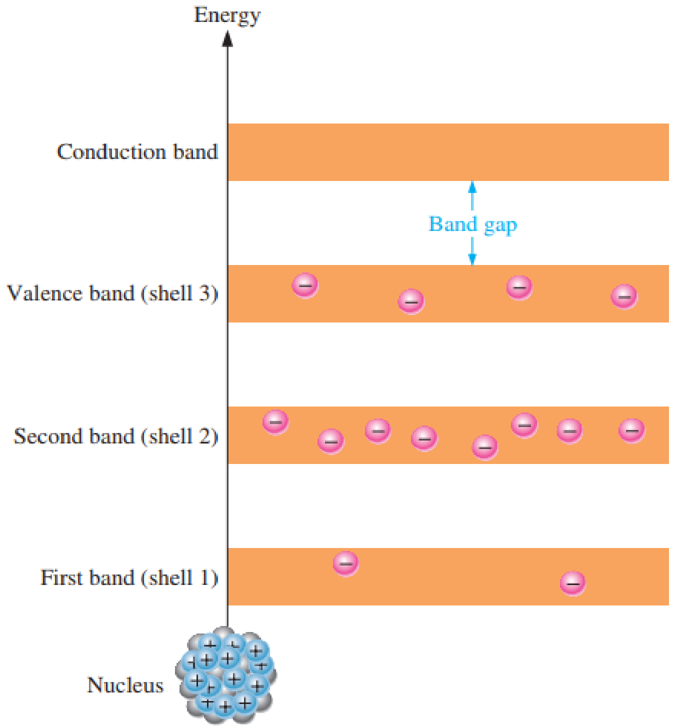


Fig 1.12 — Energy bands of intrinsic silicon at 0 K

Conduction Electrons & Holes

- **Creating a Free Carrier**
  - Heat (or light) gives a valence electron enough energy to cross the band gap
  - It enters the conduction band as a conduction (free) electron — now mobile
  - The empty spot it leaves behind in the valence band is a hole
- **The Electron–Hole Pair**
  - Every electron promoted creates exactly one hole — they appear in pairs
  - The free electron carries negative charge; the hole acts like a positive vacancy
  - Two kinds of carrier now exist in the crystal

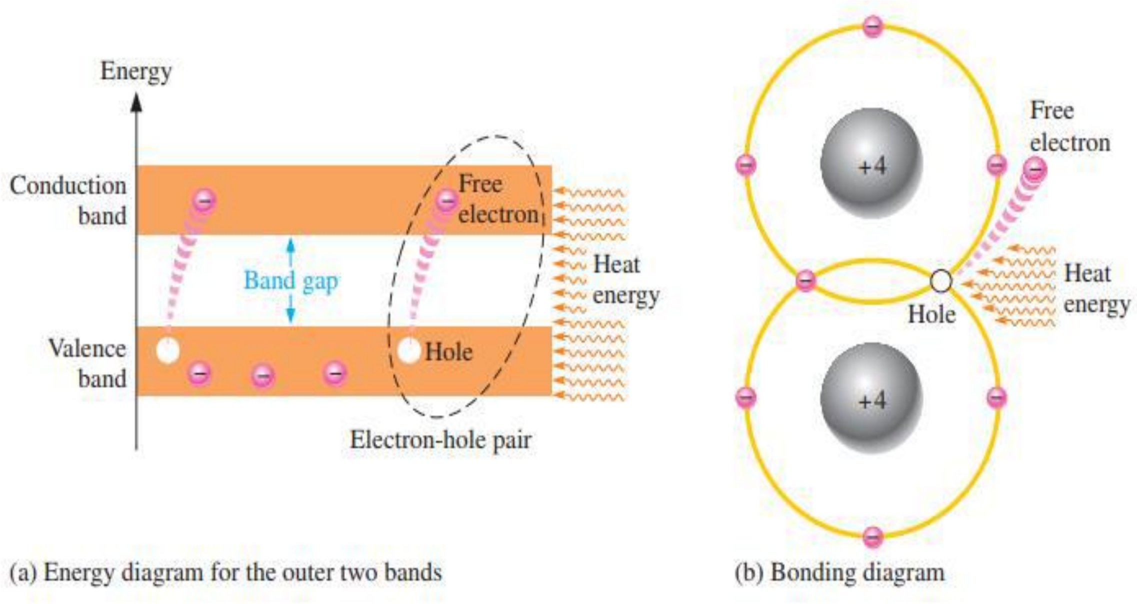


Fig 1.13 — Creating an electron–hole pair: (a) energy diagram, (b) bonding diagram

Generation & Recombination

- **Two Competing Processes**
  - Generation — thermal energy continually breaks bonds, creating new electron–hole pairs
  - Recombination — a free electron loses energy and falls back into a hole, and the pair disappears
  - At a fixed temperature the two rates balance, so the number of carriers is steady

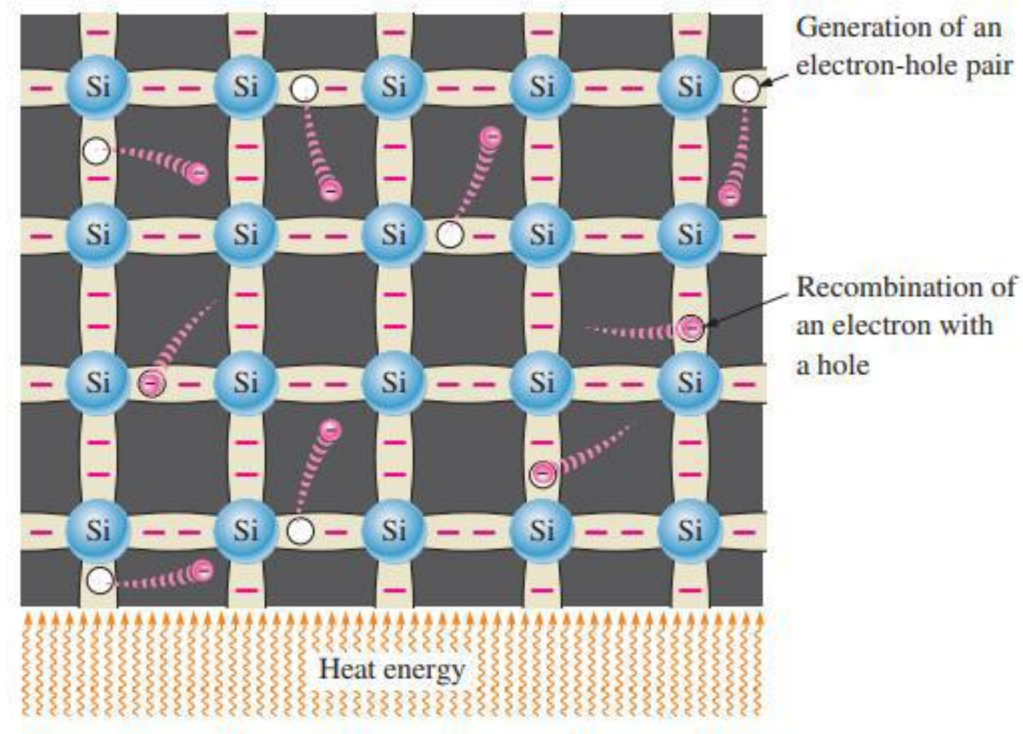


Fig 1.14 — Electron–hole pairs: continuous generation and recombination

Electron Current

- **How It Flows**
  - Connect a voltage source across the silicon
  - Conduction-band electrons are free to move, so they drift toward the positive end of the material
  - This flow of free electrons through the conduction band is electron current
- **Connecting Back**
  - These are the same free electrons we just created by breaking bonds
  - In pure silicon there are only a few, so the current is small

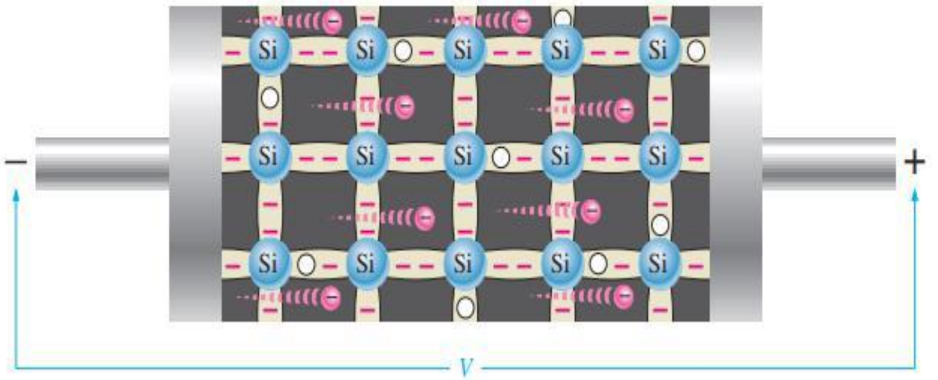


Fig 1.15 — Electron current in intrinsic silicon

Hole Current

- **How a Hole Moves**
  - A valence electron next to a hole can jump into it — without entering the conduction band
  - Filling one hole leaves a new hole where that electron came from
  - Repeated hopping makes the hole appear to travel through the valence band
- **The Key Idea**
  - The hole moves in the opposite direction to the hopping valence electrons
  - This valence-band flow is hole current

Hole Current Visualized

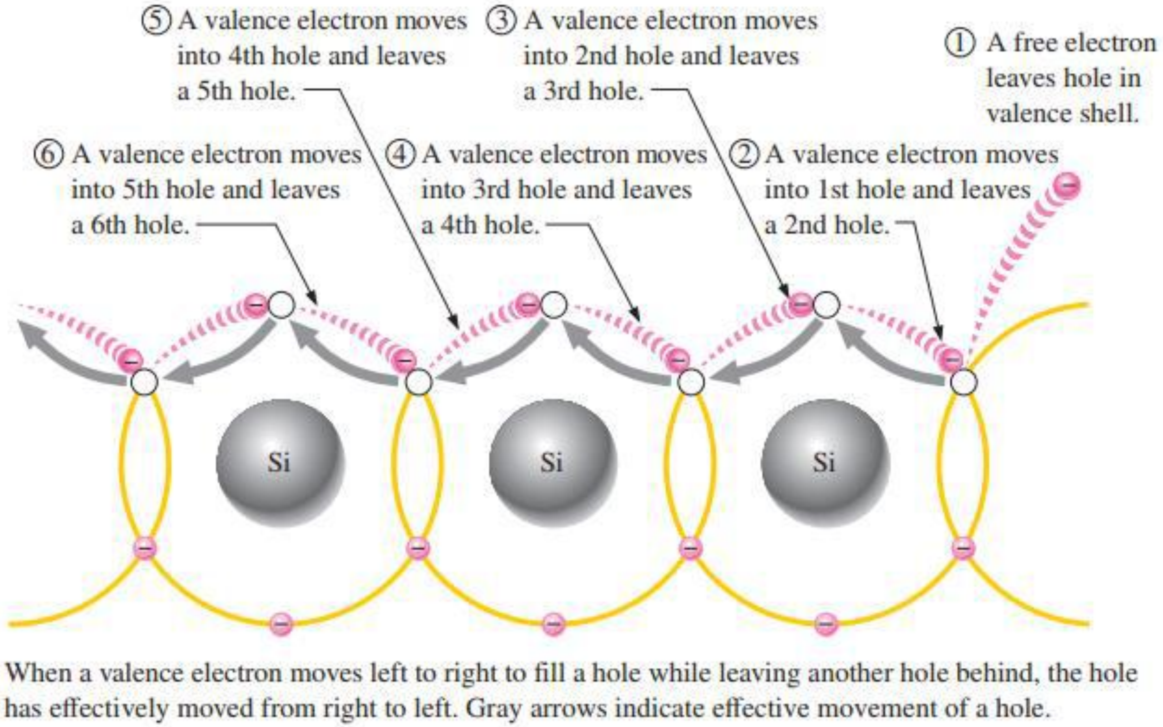


Fig 1.16 — Hole current in intrinsic silicon

Carriers in Pure Silicon & the Role of Temperature

- **A Small, Steady Supply**
  - At a fixed temperature, generation and recombination balance, so the number of carriers holds steady
  - In pure silicon, carriers are always made in pairs — one free electron and one hole together
  - At room temperature only a tiny fraction of the bonds are broken — far too few carriers for good conduction
- **Why Temperature Matters**
  - Heat the crystal and many more bonds break → many more carriers
  - So a pure semiconductor conducts BETTER when heated
  - This strong temperature sensitivity is exactly why we dope: to set the carrier count by design, not by temperature

Two Currents in a Semiconductor — and the Contrast with a Metal

- **Semiconductor (e.g. silicon)**
  - Built from covalent bonds — carriers appear only when a bond is broken
  - Two carrier types: free electrons (conduction band) and holes (valence band)
  - Carrier supply depends on temperature — very few at room temperature
- **Metal (e.g. copper)**
  - Each atom gives up its single valence electron to a shared “sea” that belongs to the whole crystal
  - Left behind are fixed positive ion cores; the attraction between them and the electron sea is the metallic bond
  - Those electrons are always free and hugely numerous → an excellent conductor even at room temperature
  - Only ONE carrier type — electrons. There are no holes: with no bonds there are no vacancies to move, and the cores are locked in place
- **Why It Matters**
  - Holes exist only in semiconductors — and the hole is what makes the diode possible

Quick Check #1 — Current in Semiconductors

- **Question 1**
  - Free (conduction) electrons exist in which energy band?
- **Question 2**
  - Which electrons are responsible for electron current?
- **Question 3**
  - What exactly is a hole, and in which band does hole current occur?
- **Question 4**
  - What happens to an electron–hole pair during recombination?
- **Question 5**
  - Name one key difference between conduction in silicon and in copper.

Part 2 — N-Type & P-Type Semiconductors

Why Dope? From Pure to Engineered Silicon

- **The Problem with Intrinsic Silicon**
  - At room temperature only about 1 atom in 10^{12} gives up an electron
  - That is far too few carriers for a useful device
  - We need a controllable way to add carriers
- **The Solution: Doping**
  - Doping = the deliberate, controlled addition of impurity atoms to a pure semiconductor
  - It increases the number of carriers and raises conductivity
- **Intrinsic vs Extrinsic**
  - Intrinsic = a pure semiconductor (carriers only from thermally broken bonds)
  - Extrinsic = a doped semiconductor (impurity atoms supply most of the carriers)

Doping: The Big Idea

- **What Doping Does**
  - A few silicon atoms are replaced by impurity atoms in the crystal lattice
  - The impurity adds carriers of one chosen type
  - Adjusting how many impurity atoms we add controls how many carriers we get
- **Two Choices of Dopant**
  - Pentavalent (5 valence electrons) → adds a free electron → n-type
  - Trivalent (3 valence electrons) → creates a hole → p-type

TABLE 1 Portion of the Periodic Table Related to Semiconductors

Period	Column II	III	IV	V	VI
2		B Boron	C Carbon	N Nitrogen	O Oxygen
3	Mg Magnesium	Al Aluminum	Si Silicon	P Phosphorus	S Sulfur
4	Zn Zinc	Ga Gallium	Ge Germanium	As Arsenic	Se Selenium
5	Cd Cadmium	In Indium	Sn Tin	Sb Antimony	Te Tellurium
6	Hg Mercury		Pb Lead		

How Little Dopant It Takes

- **A Remarkably Small Amount**
  - Only about one silicon atom in several million is replaced by a dopant atom
  - Everything else is still ordinary silicon
- **The Insight**
  - That tiny, controlled trace of dopant completely sets how the silicon conducts — this is the power of doping

N-Type Semiconductor: Pentavalent Donor Atoms

- **Pentavalent (Donor) Atoms**
  - Impurity atoms with five valence electrons — e.g., arsenic (As), phosphorus (P), bismuth (Bi), antimony (Sb)
  - Four of the five electrons form covalent bonds with neighbouring silicon atoms
  - The fifth electron has no bond to join — it becomes a free conduction electron
  - Because it gives away an electron, the atom is called a donor atom
- **Important**
  - No hole is created — doping n-type adds only a free electron
  - The number of conduction electrons can be carefully controlled by the number of impurity atoms added to the silicon

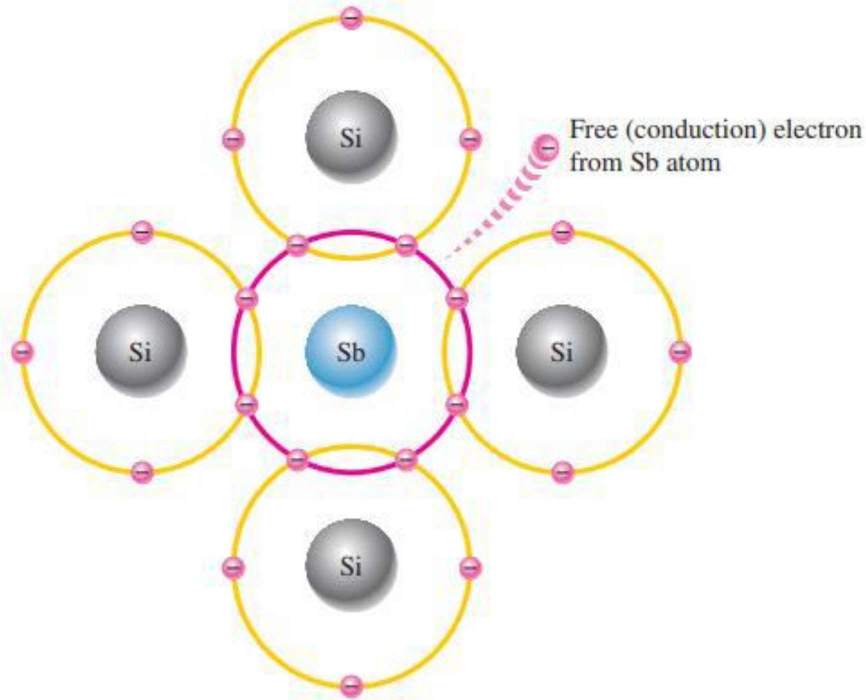


Fig 1.17 — Pentavalent (donor) atom: antimony (Sb)

N-Type Semiconductor: Majority & Minority Carriers

- **Majority Carrier — Electrons**
  - Each donor atom adds one free electron
  - Free electrons greatly outnumber holes
  - “n-type” is named for the negative charge of the electron
- **Minority Carrier — Holes**
  - A few electron–hole pairs are still produced thermally
  - The holes from those pairs are the minority carrier — far fewer than the electrons
- **Remember**
  - Doping sets the (large) majority count; temperature sets the (small) minority count

P-Type Semiconductor: Trivalent Acceptor Atoms

- **Trivalent (Acceptor) Atoms**
  - Impurity atoms with three valence electrons — e.g., boron (B), indium (In), gallium (Ga)
  - All three electrons form covalent bonds with neighbouring silicon atoms
  - One bond position is left empty — that vacancy is a hole
  - Because it can take in an electron, the atom is called an acceptor atom
- **Important**
  - No free electron is added — doping p-type creates only a hole

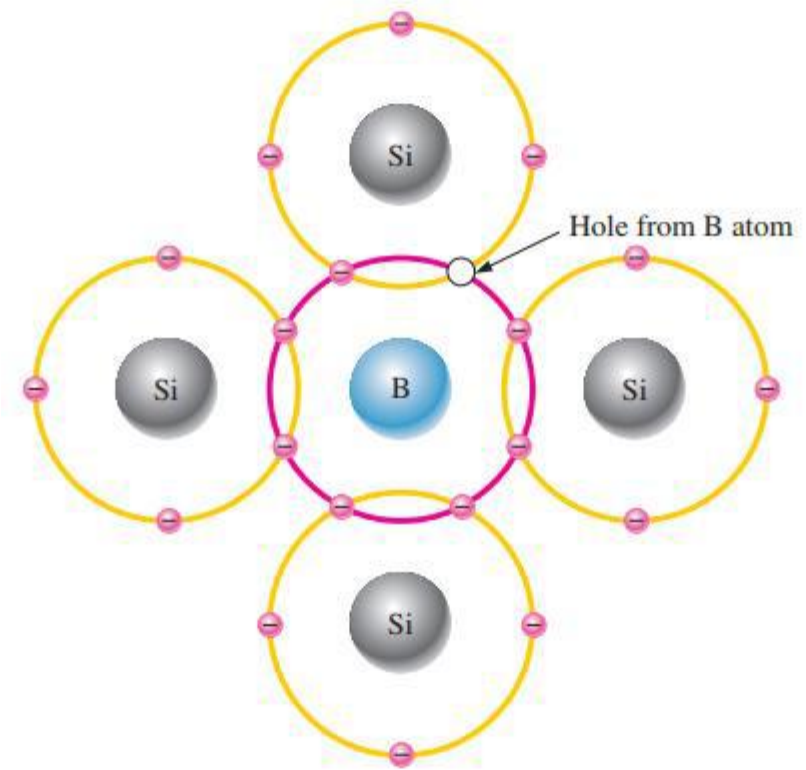


Fig 1.18 — Trivalent (acceptor) atom: boron (B)

P-Type Semiconductor: Majority & Minority Carriers

- **Majority Carrier — Holes**
  - Each acceptor atom creates one hole
  - Holes greatly outnumber free electrons
  - “p-type” is named for the positive nature of the hole
- **Minority Carrier — Electrons**
  - A few electron–hole pairs still form thermally
  - The electrons from those pairs are the minority carrier — far fewer than the holes
- **Symmetry**
  - p-type is the mirror image of n-type — swap “electron” and “hole” throughout

Majority and Minority Carriers — Why One Dominates

- **Two Populations in Every Doped Crystal**
  - Doping fills the crystal with one chosen carrier — the majority carrier (electrons in n-type, holes in p-type)
  - Thermal energy still makes a few electron–hole pairs, giving a small number of the opposite carrier — the minority carrier
- **Why the Minority Is So Scarce**
  - With majority carriers everywhere, any minority carrier is very likely to meet one and recombine
  - So the more heavily you dope, the more the majority dominates and the fewer minority carriers survive
- **Tie It Back**
  - n-type: many electrons (majority), very few holes (minority)
  - p-type: many holes (majority), very few electrons (minority)

N-Type vs P-Type — Side by Side

- **N-Type**
  - Dopant: pentavalent / donor — As, P, Bi, Sb (5 valence electrons)
  - Adds a free electron (no hole)
  - Majority = electrons · Minority = holes
- **P-Type**
  - Dopant: trivalent / acceptor — B, In, Ga (3 valence electrons)
  - Creates a hole (no free electron)
  - Majority = holes · Minority = electrons
- **Common Confusion**
  - Doped silicon is still electrically neutral — adding carriers does not add net charge; every atom is itself neutral

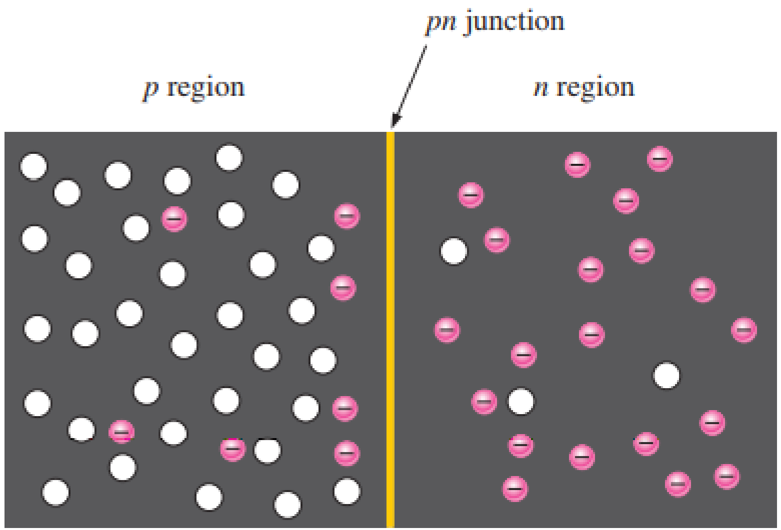
Quick Check #2 — Doping, N-Type & P-Type

- **Question 1**
  - Define doping in one sentence.
- **Question 2**
  - Pentavalent vs. trivalent — how many valence electrons each, and what is each also called?
- **Question 3**
  - How is n-type material formed? How is p-type material formed?
- **Question 4**
  - State the majority and minority carrier for n-type and for p-type.
- **Question 5**
  - What is the difference between intrinsic and extrinsic silicon?

Part 3 — The PN Junction

Forming a PN Junction

- **How it is Built**
  - One block of silicon: a p-type region on one side, an n-type region on the other
  - The boundary between them is the pn junction
  - This simple structure is a diode — and the basis of transistors and solar cells
- **Important**
  - p region: holes are the majority, a few electrons are the minority
  - n region: electrons are the majority, a few holes are the minority
  - Each region, on its own, is electrically neutral

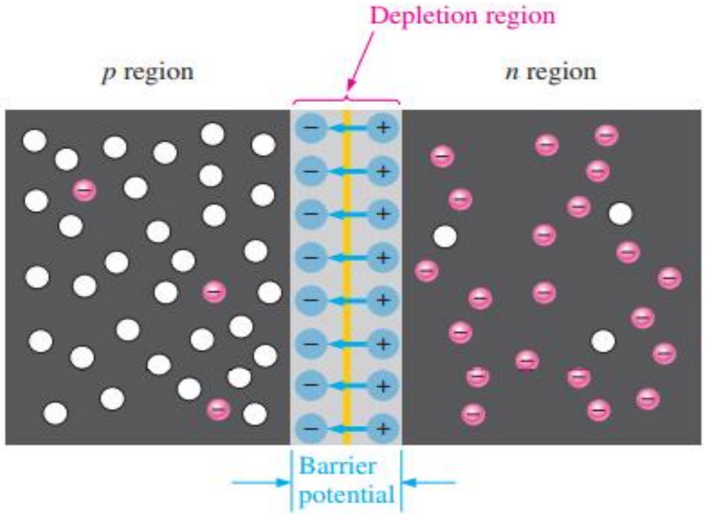


(a) The basic silicon structure at the instant of junction formation showing only the majority and minority carriers. Free electrons in the *n* region near the *pn* junction begin to diffuse across the junction and fall into holes near the junction in the *p* region.

Fig 1.19(a) — The pn junction at formation

Diffusion & the Depletion Region

- **Diffusion at the Junction**
  - Free electrons near the junction diffuse from the n region into the p region
  - There they recombine with holes
  - Each departing electron leaves behind a fixed positive (pentavalent) ion on the n side
  - Each filled hole leaves a fixed negative (trivalent) ion on the p side
- **The Depletion Region**
  - These exposed ions form two layers of opposite charge at the junction
  - This thin layer, now emptied of free carriers, is the depletion region
  - It forms very quickly and is very narrow



(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field.

Fig 1.19(b) — Formation of the depletion region & barrier potential

Equilibrium — Why Diffusion Stops

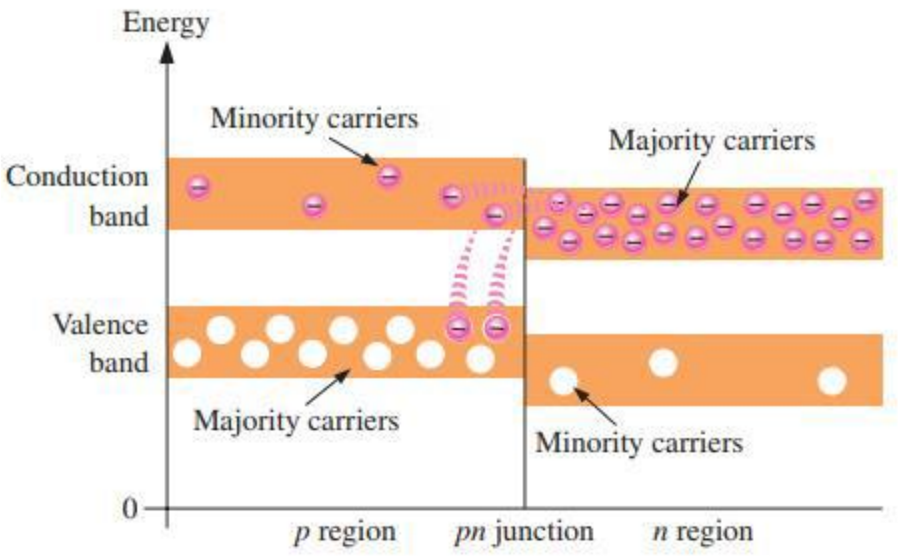
- **The Self-Limiting Process**
  - Step 1 — n-side electrons cross the junction and fill p-side holes
  - Step 2 — fixed negative ions build up on the p side of the junction
  - Step 3 — that negative charge repels further electrons (like charges repel), so diffusion halts
- **Result**
  - The depletion region now acts as a barrier to further crossing

The Barrier Potential

- **What Creates It**
  - The two layers of opposite charge create an electric field across the junction (Coulomb's law)
  - Energy is needed to move an electron across that field
  - The voltage equivalent of that energy is the barrier potential
- **Typical Values (at 25 °C)**
  - Silicon (Si): about 0.7 V
  - Germanium (Ge): about 0.3 V
- **What It Depends On**
  - The type of semiconductor material, the amount of doping, and the temperature

Energy Diagram (1): At Formation

- **Why the Offset**
  - A trivalent atom holds its electrons a little more loosely than a pentavalent one
  - So the p-region energy bands sit at slightly higher energy than the n-region bands
- **At the Instant of Formation**
  - The n-region and p-region bands overlap in energy
  - High-energy electrons in the n-side conduction band can diffuse across…
  - …and drop down into holes in the p-side valence band



(a) At the instant of junction formation

Fig 1.20(a) — Energy diagram at the instant of junction formation

Energy Diagram (2): Equilibrium & the Energy Hill

- **Reaching Equilibrium**
  - As electrons leave, the n-region bands shift downward in energy
  - This continues until the top of the n-side conduction band lines up with the bottom of the p-side conduction band
  - Now an n-side electron needs extra energy to cross — so diffusion stops
- **The Energy Hill**
  - This energy gradient at the junction is the energy hill
  - The valence band shifts down by the same amount — so each material's band gap is unchanged

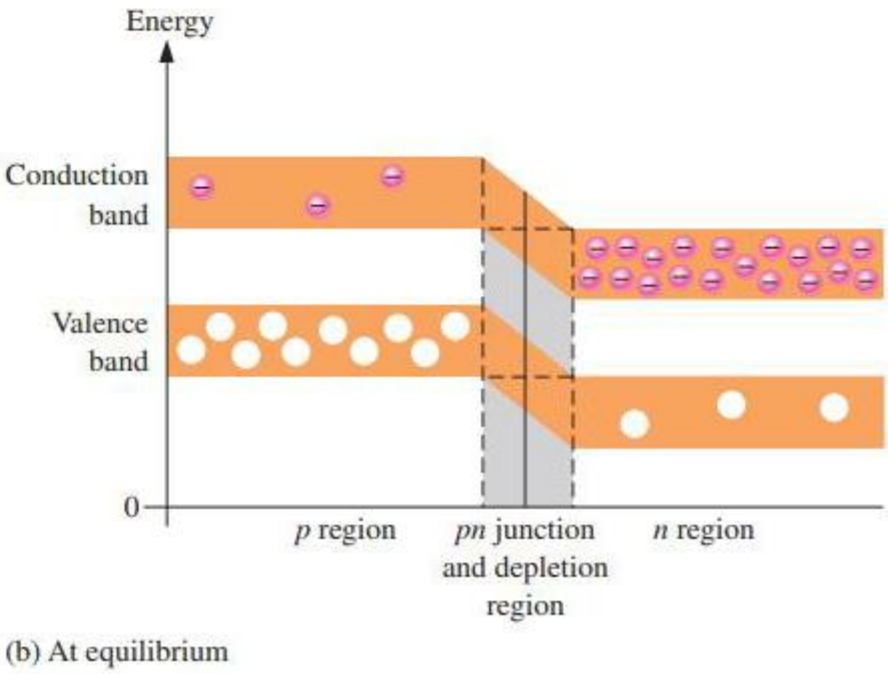


Fig 1.20(b) — At equilibrium: the energy hill

Quick Check #3 — The PN Junction

- **Question 1**
  - What is a pn junction?
- **Question 2**
  - Explain what happens during diffusion at the junction.
- **Question 3**
  - Describe the depletion region — what is in it, and what is not?
- **Question 4**
  - What is the barrier potential, and how is it created?
- **Question 5**
  - Typical barrier potential for silicon? For germanium?

Part 4 — Wrap-Up

Key Takeaways from Today

- Current in a semiconductor needs free carriers — at 0 K there are none
- Breaking a bond creates an electron–hole pair; generation and recombination stay in balance
- Semiconductors carry current two ways — electron current (conduction band) and hole current (valence band); a metal uses only free electrons
- Doping adds carriers by design: pentavalent donors → n-type (electrons majority); trivalent acceptors → p-type (holes majority)
- Doped silicon is still electrically neutral
- At a pn junction, diffusion builds a depletion region and a barrier potential (~0.7 V Si, ~0.3 V Ge)
- On the energy diagram, the n-region bands shift down to form the energy hill at equilibrium

Coming Up — Week 3: Diode Operation

- **Next Topic — Diode Operation (Lecture Notes pp. 28–46)**
  - Forward bias — overcoming the barrier potential so the diode conducts
  - Reverse bias — widening the depletion region so it (almost) does not
  - The current–voltage (I–V) behaviour of a real diode
  - Diode models in circuits

Thank You — Questions?